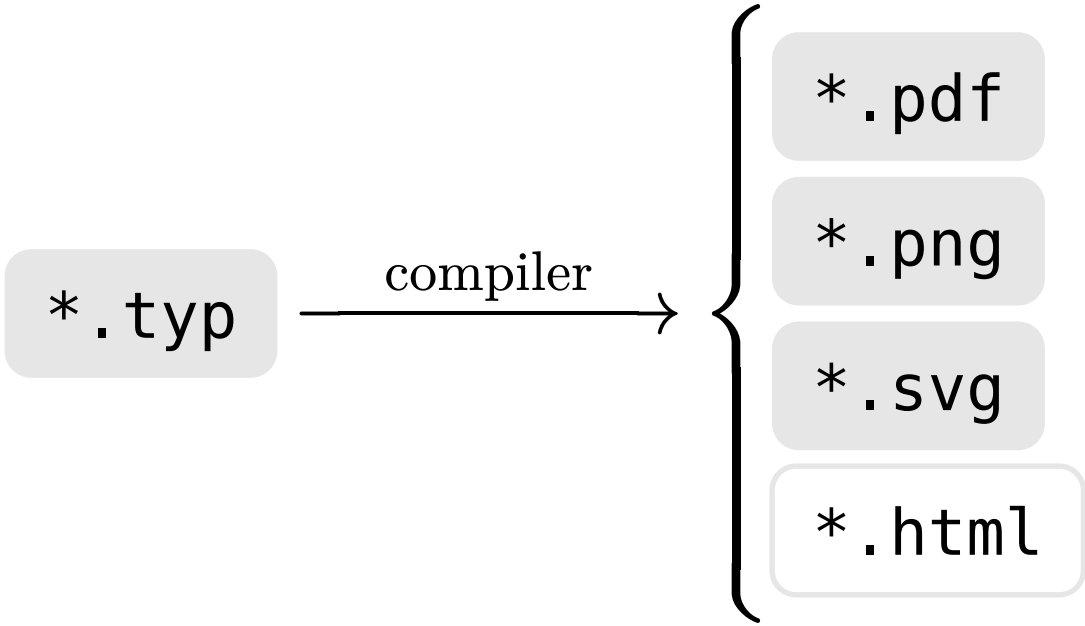
AN *INTRODUCTION* TO

typst

Typst is a
typesetting
system


*.typ
compiler
*.pdf
*.png
*.svg
*.html

Typst is a
markup
language

3 modes:

markup, *math*, code

Markup mode

```
Odi et amo.
Quare id faciam fortasse requiris.\
Nescio,
sed fieri sentior et excrucior.

Catullus
```

Odi et amo. Quare id faciam fortasse requiris.
Nescio, sed fieri sentior et excrucior.

Catullus

```
We can make our text *bold* or
_italic_.
```

We can make our text **bold** or *italic*.

```
= Big title
== Small title
=== Smaller title
==== Even smaller title
```

Big title

Small title

Smaller title

Even smaller title

```
Three italian dishes are:
- Pasta
- Pizza
- Tiramisù
```

Three italian dishes are:

- Pasta
- Pizza
- Tiramisù

```
My favorite mangas are:
+ One Piece
+ Attack on Titan
```

My favorite mangas are:

1. One Piece
2. Attack on Titan

```
We can write code inline
like this `print("Hello")` or
like this ```py print("Hello")```.
```

We can write code inline like this `print("Hello")` or like this `print("Hello")`.

```
We can make a code block like this:
```py
print("Hello")
```
```

We can make a code block like this:

```py
print("Hello")
```

```
// We can make a comment
/* In two different ways */
```

Math mode

```
The most beautiful equation in math is considered by many to be $e^(i pi) = -1$.
```

The most beautiful equation in math is considered by many to be $e^{i\pi} = -1$.

```
A more general formula exists:
$
e^(i theta)
= sum_(n=0)^(+oo) (i theta)^n/(n!)
= cos(theta) + i sin(theta).
$
```

A more general formula exists:

$$e^{i\theta} = \sum_{n=0}^{+\infty} \frac{(i\theta)^n}{n!} = \cos(\theta) + i\sin(\theta).$$

```
$
A inter B = emptyset =>
A without B = A
$
```

$$A \cap B = \emptyset \Rightarrow A \setminus B = A$$

```
$
Q = 2 pi
"energy at the start" /
"energy dissipated"
$
```

$$Q = 2\pi \frac{\text{energy at the start}}{\text{energy dissipated}}$$

```
$ arrow(F) = m arrow(a) $
```

$$\vec{F} = m\vec{a}$$

```
$
vec(3,9,15) =
mat(1,2,3; 4,5,6; 7,8,9)
vec(1,1,0)
$
```

$$\begin{pmatrix} 3 \\ 9 \\ 15 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$$

```
$
f(x) &= (x+1)^2 &&= x^2 + 2x + 1 \
g(x) &= 10(x-1) &&= 10x - 10 \
$
```

$$\begin{aligned} f(x) &= (x+1)^2 &&= x^2 + 2x + 1 \\ g(x) &= 10(x-1) &&= 10x - 10 \end{aligned}$$

Code mode

```
#bibliography("example.bib")
```

Bibliography

[1] M. A. N. & Isaac L. Chuang, *Quantum Computation and Quantum Information.* Cambridge University Press, 2010.

[2] P. W. Shor, “Scheme for reducing decoherence in quantum computer memory,” *Physical review A*, vol. 52, no. 4, p. R2493, 1995.

```
#strong("bold"), *bold*
```

bold, **bold**

```
#link("example.com", "Example")
```

Example

```
#let x = "hello"
I want to say you #x
```

I want to say you hello

```
#let x = [Typst is _amazing_]
#let y = "Typst is _amazing_"
#x \ #y
```

Typst is *amazing*
Typst is _amazing_

```
#let nice(it) = [✨] + it + [✨]
#nice([Typst]) \ #nice[Typst]
```

✨Typst✨
✨Typst✨

```
#link("example.com",[_Example_])
#link("example.com")[_Example_]
```

Example Example

```
#figure(
  image("cow.jpg", width: 6cm),
  caption: [Cowdynamics],
)
```

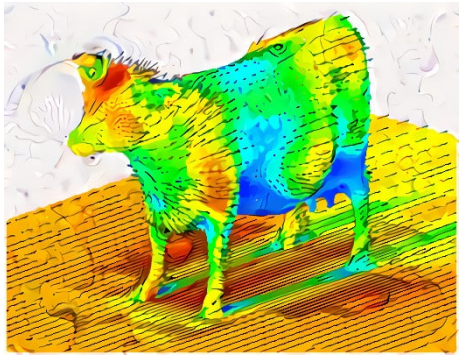


Figure 1: Cowdynamics

```
#{
  let N = 10
  let n = 1
  [1 ]
  for i in range(2, N + 1) {
    n += i
    [\+ #i ]
  }
  [\= #n]
}
```

$$1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 = 55$$

```
#{
  let bool = true
  let list = (1, 2, 3)
  let dict = (a: 1, b: 2)
  let lambda = x => x * 2
}
```

set & show

element

```
#set text(size: 30pt)
```

new defaults

```
#set text(
  font: "New Computer Modern"
)
Hello Typst enthusiasts!
```

Hello Typst enthusiasts!

```
#set heading(numbering: "1.i")

= A primary heading
The first block of text after the
title

== A secondary heading
The paragraph right after the
secondary heading
```

1 A primary heading

The first block of text after the title

1.i A secondary heading

The paragraph right after the secondary heading

```
#set heading(numbering: "1")

= Referenceable <ref>
This is @ref
```

2 Referenceable

This is Section 2

```
#set list(marker: "👉", indent: 1cm)

- Element A
  - Element A1
  - Element A2
- Element B
```

- 👉 Element A
  - 👉 Element A1
  - 👉 Element A2
- 👉 Element B

```
#set page(margin: (
  top: 2cm,
  bottom: 3cm,
  left: 3cm,
  right: 3cm,
))
```

```
#{
  set text(fill: red)
  [Alert!]
}
Typst is coming!
```

Alert! Typst is coming!

$$\texttt{\#show}\ \overbrace{\texttt{link}}^{\text{filter}}\texttt{:}\ \underbrace{\texttt{underline}}_{\text{editing function}}$$

```
#show link: underline
```

means

$$(\texttt{link} \mapsto \texttt{underline} \circ \texttt{link})$$

```
#show link: underline
#link("example.com", "Example")
```

Example

```
#show heading: it => {
  set text(
    size: 30pt,
    font: "Helvetica"
  )
  it
}

= Lorem ipsum
#lorem(30)
```

Lorem ipsum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.

```
#show heading: set text(
  size: 30pt,
  font: "Helvetica"
)

= Lorem ipsum
#lorem(30)
```

Lorem ipsum

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri.

```
#show math.equation: align.with(left)
$ F = m a $
```

$$F = ma$$

Templates

```
#import "@preview/charged-ieee:0.1.3": ieee

#show: ieee.with(
  title: [A Typesetting System to ..],
  abstract: [..],
  authors: ((name: "Martin Haug", ..), ..),
  bibliography: bibliography("example.bib"),
  ..
)

= Introduction
..
```

A Typesetting System to Untangle the Scientific Writing Process

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***Abstract*—The process of scientific writing is often tangled up with the intricacies of typesetting, leading to frustration and wasted time for researchers. In this paper, we introduce Typst, a new typesetting system designed specifically for scientific writing. Typst untangles the typesetting process, allowing researchers to compose papers faster. In a series of experiments we demonstrate that Typst offers several advantages, including faster document creation, simplified syntax, and increased ease-of-use.**

***Index Terms*—Scientific writing, Typesetting, Document creation, Syntax**

I. Introduction

Scientific writing is a crucial part of the research process, allowing researchers to share their findings with the wider scientific community. However, the process of typesetting scientific documents can often be a frustrating and time-consuming affair, particularly when using outdated tools such as LaTeX. Despite being over 30 years old, it remains a popular choice for scientific writing due to its power and flexibility. and provide evidence for its superiority over other typesetting systems in a variety of scenarios.

II. Methods

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem.

$$a + b = \gamma \tag{1}$$

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non

Compilation

LOCALLY

- `typst compile example.typ`
- `typst watch example.typ`

ONLINE

- https://typst.app/

Packages

```
#import "@preview/physica:0.9.8": *
```

```
$
curl (grad V + pdv(vb(A), t)) =
pdv(vb(B), t)
$
```

$$\boldsymbol{\nabla} \times \left(\boldsymbol{\nabla} V + \frac{\partial \boldsymbol{A}}{\partial t} \right) = \frac{\partial \boldsymbol{B}}{\partial t}$$

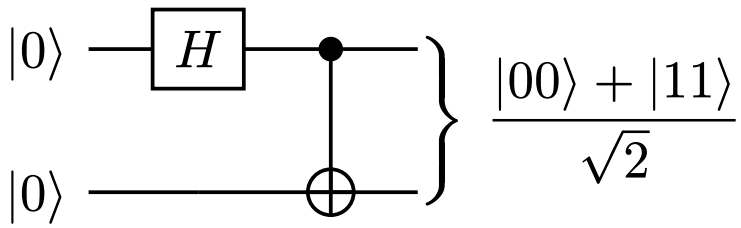
```
#let tn = tensor
$
tn(R, +mu, -nu) -
1/2 R tn(g, +mu, -nu) =
8 pi G tn(T, +mu, -nu)
$
```

$$R^{\mu}{}_{\nu} - \frac{1}{2} R g^{\mu}{}_{\nu} = 8\pi G T^{\mu}{}_{\nu}$$

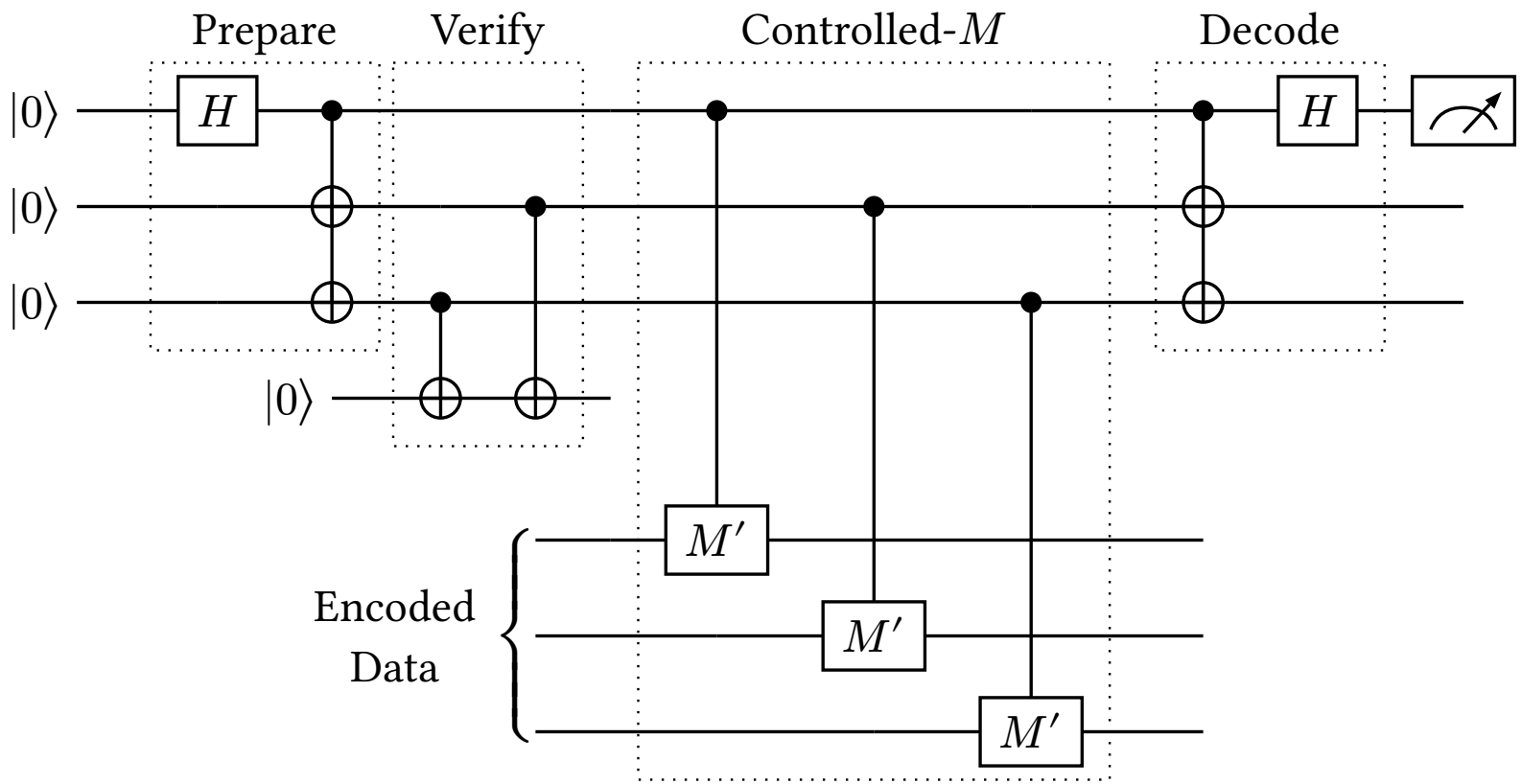
#import "@preview/quill:0.7.2": *

```
#import "@preview/quill:0.7.2": *
#import "@preview/physica:0.9.8": ket

#let super = $ (ket(00) + ket(11))/sqrt(2) $
#quantum-circuit(
  lstick($ket(0)$), $H$, ctrl(1), rstick(super, n: 2), [\ ],
  lstick($ket(0)$), 1, targ(), 1
)
```

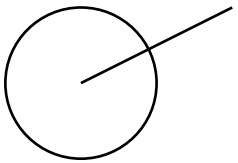


https://github.com/Mc-Zen/quill/blob/v0.6.1/examples/fault-tolerant-measurement.typ



```
#import "@preview/cetz:0.4.2"
```

```
#import "@preview/cetz:0.4.2"
#cetz.canvas({
  import cetz.draw: *
  circle((0, 0))
  line((0, 0), (2, 1))
})
```



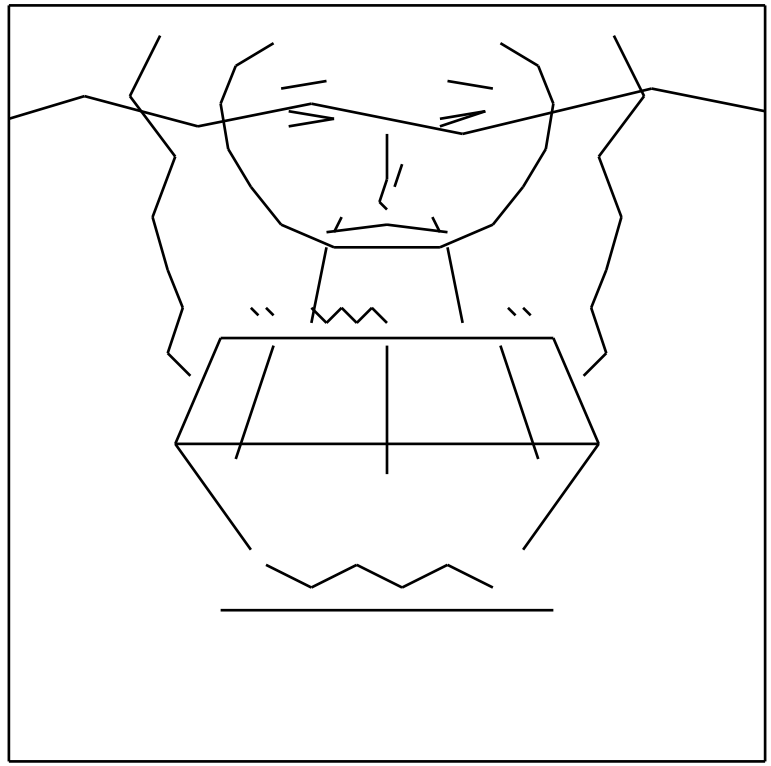
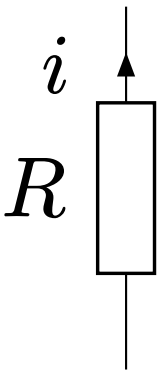


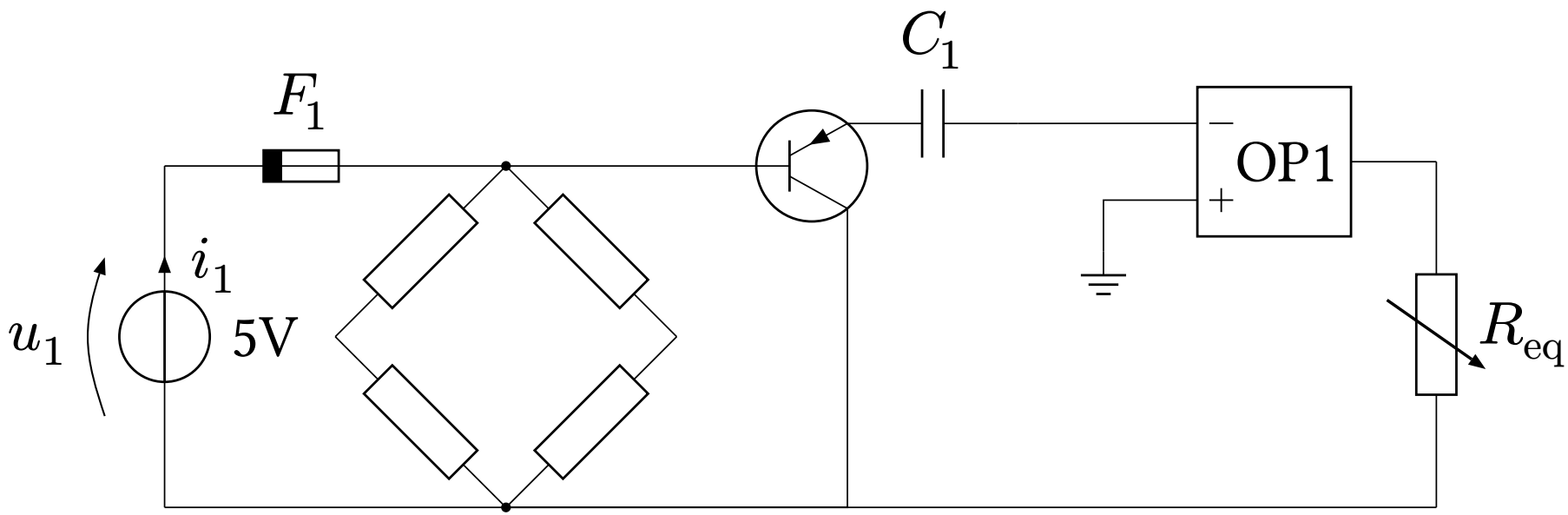
Figure 2: *La Mona Lisa* according to chatGPT

#import "@preview/zap:0.5.0"

```
#import "@preview/zap:0.5.0"
#zap.circuit({
    import zap: *
    resistor("r1", (0,0), (0, 3),
      label: $R$, i: $i$
    )
})
```



https://github.com/l0uisgrange/zap/blob/main/examples/example1.typ



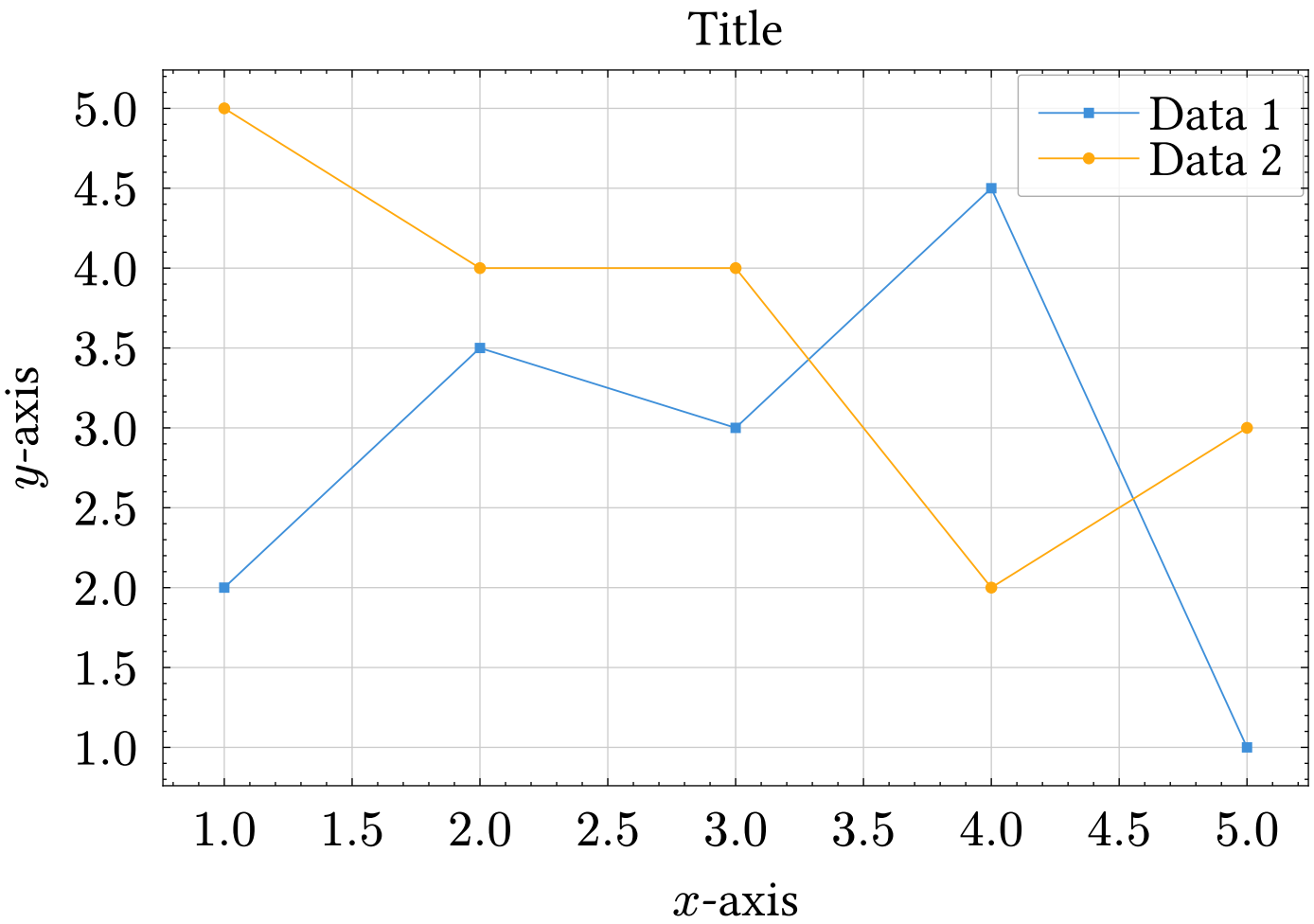
```
#import "@preview/lilaq:0.6.0" as lq
```

```
#let data = csv("example.csv").slice(1, none)
#let datax = data.map(row => float(row.at(0)))
#let datay1 = data.map(row => float(row.at(1)))
#let datay2 = data.map(row => float(row.at(2)))

#lq.diagram(
  title: [Title],
  xlabel: [$x$-axis],
  ylabel: [$y$-axis],
  width: 16cm,
  height: 10cm,
  lq.plot(datax, datay1, mark: "s", label: [Data 1]),
  lq.plot(datax, datay2, mark: "o", label: [Data 2])
)
```

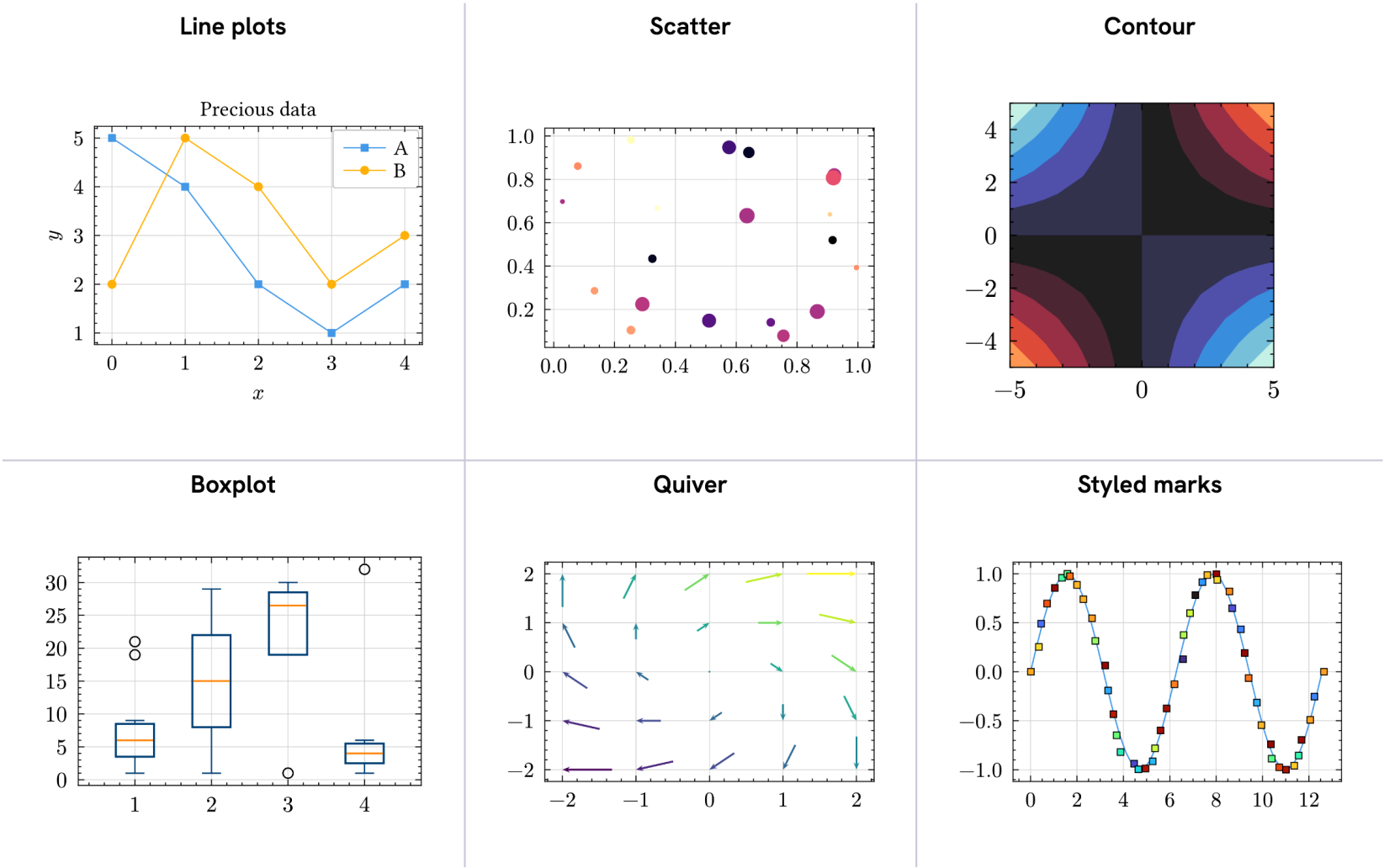
example.csv

datax	datay1	datay2
1	2	5
2	3.5	4
3	3	4
4	4.5	2
5	1	3



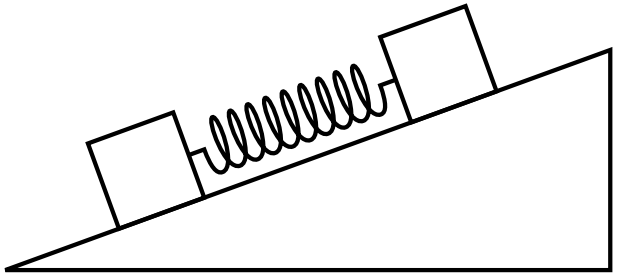
example.csv

datax	datay1	datay2
1	2	5
2	3.5	4
3	3	4
4	4.5	2
5	1	3

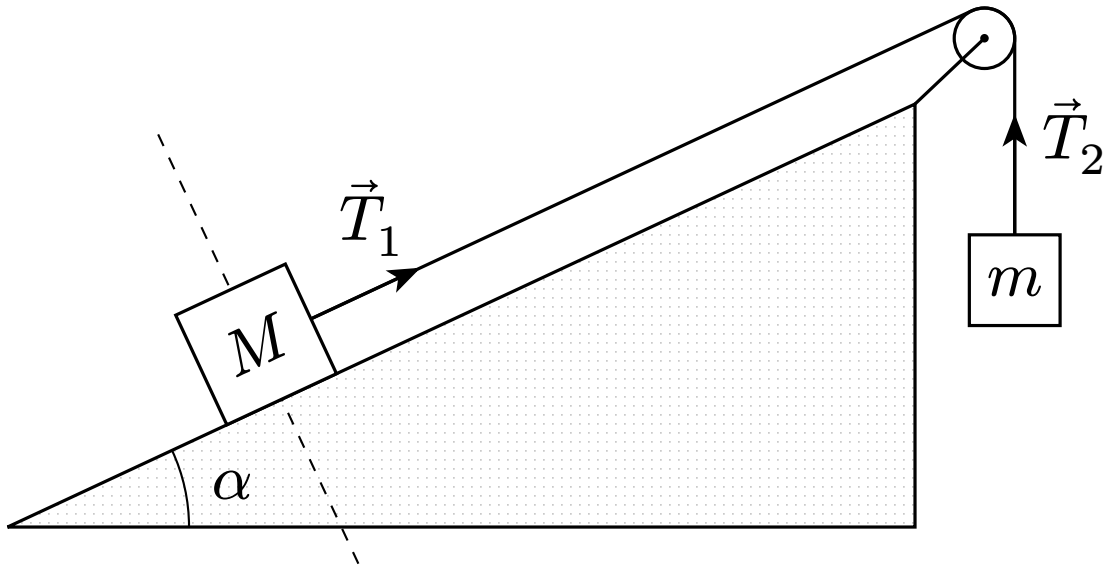

Line plots
Precious data
A
B
y
x
Scatter
Contour
Boxplot
Quiver
Styled marks

#import "@preview/patatrac:0.5.0"

```
#import "@preview/patatrac:0.5.0"
#patatrac.cetz.canvas({
  import patatrac: *
  let draw = cetz.standard()
  let floor = incline(100, 20deg)
  let A = rect(15, 15)
  let B = rect(15, 15)
  A = stick(A("bl"), floor("tl"))
  B = stick(B("br"), floor("tr"))
  A = slide(A("c"), +20, 0)
  B = slide(B("c"), -20, 0)
  let k = spring(A("r"), B("l"))
  draw(floor, A, B, k)
})
```



https://github.com/ZaninDavide/patatrac/blob/main/gallery/incline.typ



#import "@preview/beam:0.1.1"

```
#import "@preview/beam:0.1.1"
#beam.setup({
  import beam: *
  laser("laser", (0, 0))
  lens("l1", (3, 0))
  detector("cam", (4, 0))
  beam("", "laser", "l1")
  focus("", "l1", "cam")
})
```

```
#import "@preview/wicked:0.2.0": wick
```

```
$ :
wick(id: #1, macron(Psi))_alpha (x)
gamma^mu_(alpha beta)
wick(pos: #top, A)_mu (x)
wick(Psi)_beta (x)
wick(macron(Psi))_eta (y)
gamma^nu_(eta rho) (y)
wick(pos: #top, A)_nu
wick(id: #1, Psi)_rho (y)
: $
```

$$: \bar{\Psi}_\alpha(x)\gamma^\mu_{\alpha\beta}A_\mu(x)\Psi_\beta(x)\bar{\Psi}_\eta(y)\gamma^\nu_{\eta\rho}(y)A_\nu\Psi_\rho(y) :$$

&context
&introspection

```
#context { text.size },
#context { text.font }
```

20pt, libertinus serif

```
Introduction is at:\
#context {
  locate(<intro>).position()
}

= Introduction <intro>
```

Introduction is at:
(page: 46, x: 430.94pt, y: 211.95pt)

Introduction

```
#context {
  for h in query(
    heading.where(level: 1)
  ) {
    h.body
    linebreak()
  }
}
```

Big title
Bibliography
A primary heading
Referenceable
Lorem ipsum
Lorem ipsum
Introduction

STRANGER THINGS

boxes

```
This is a
#box(
  fill: red, radius: 10pt,
  inset: (x: 10pt), outset: (y: 10pt),
  text(fill: white)[`BOX`]
)

This is a
#block(fill: blue, radius: 10pt, outset: 10pt,
  text(fill: white)[`BLOCK`]
)
```

This is a BOX

This is a

BLOCK

2D input in math mode

```
#let func(..args) = repr(args)
$ func(1,2,3; 4,5,6) $
```

$$\text{arguments}(([1], [2], [3]), ([4], [5], [6]))$$

metadata

```
Metadata is invisible content:
#metadata("Useful information")

That we can query: #context {
  query(metadata).last().value
}
```

Metadata is invisible content:

That we can query: Useful information

WASM plugins

```
#let square(n) = {
  import plugin("example.wasm"): square
  int.from-bytes(square(bytes((n,))))
}
#square(5)
```

25

Find this presentation

```
#import "@preview/tiaoma:0.3.0": barcode
#barcode("https://github.com/ZaninDavide/typst-intro", "QRCode")
```

https://github.com/ZaninDavide/typst-intro